Safaan Sadiq

Oak Brook, IL | P: +1 (630) 806-5615 | sadiq.safaan@gmail.com |Linkedin: www.linkedin.com/in/safaan-sadiq |
Github: https://github.com/sadiq-safaan

Summary: Early career mathematician with strong problem solving skills, coming from a balanced background in pure and applied mathematics. Doctoral thesis studies neural dynamics of cortical decision-making using attractor neural networks. Experience includes machine learning in an industry setting along with academic exposure to HPC e.g. MPI and CUDA.

EDUCATION

Pennsylvania State University — University Park, PA
PhD Mathematics (Thesis Advisor: Carina Curto) — August 2025
Dissertation: Modeling bias in decision-making attractor networks

Northwestern University — Evanston, IL
MS Engineering Sciences and Applied Mathematics — December 2018

American University — Washington, D.C.
BS Mathematics & BA International Studies — May 2017
Honors: University Honors Program, Dean's Scholarship

Current Research Interests/Activity

- Doctoral research studies attractor network firing rate models for decision-making
- Aim to understand how properties of decision-making circuits shape bias towards choices
- Researching links between network connectivity and basins of attraction in nonlinear firing rate models
- Finding partitioning schemes to study attractors and dynamics of threshold linear networks

Pennsylvania State University — University Park, PA
Research Assistant / Faculty: Carina Curto — May – August 2022-2025
- Analyzed nonlinear neuronal firing rate models for two neuron networks to find relationships between parameters and dynamics
- Determined bifurcations on existence and stability of fixed points and on their basins of attraction
- Analyzed firing rate models with parameters derived from combinatorial objects and found correspondences between combinatorial properties and the existence of excitatory-inhibitory balanced states in the models

Research Experience

Pennsylvania State University — University Park, PA
Research Assistant / Faculty: Jessica Conway — May 2021 – August 2021
- Incorporated viral burst size into disease model to see how it impacted the extinction time of infection within the body
- Studied hybrid stochastic simulation algorithms for simultaneous discrete and continuous variables
- Implemented the above to produce simulation in R to model the disease dynamics and translated a simplified version of the stochastic simulation into C to increase speed of simulation

Northwestern University — Evanston, IL
Research Assistant / Faculty: David Chopp — October 2018 – December 2018
- Worked with a numerical solver package for Level Set methods
- Translated algorithms from the package for GPU processing with CUDA to maximize performance

Publications

- Sadiq, S. 2025. Modeling bias in decision-making attractor networks. *ArXiv* - (PhD dissertation)

Teaching Experience

Pennsylvania State University — University Park, PA
Instructor (Teaching Assistantship) — August – May 2020-2025

- Fully instructed multiple College Algebra courses with over 40 students
- Due to COVID-19 pandemic, gained experience teaching online classes during the 2020-2021 academic year
- Lectured and worked with students in-class and during office hours to help them master the material
- Managed administrative details for the course such as grading papers, writing solutions, maintaining course website, and various other general tasks that go into teaching a course.

Pennsylvania State University University Park, PA
Grader (Teaching Assistantship) January – May 2020+2024
- Graded two courses in combinatorics
- Graded two courses in introductory (proof based) discrete mathematics
- Coordinated with course professor to ensure accuracy of solutions and appropriate grading

Work Experience

Aginity Evanston, IL
Data Science Mentorship Program June 2017 – September 2017
- Conducted statistical analysis on large datasets using predictive analytics e.g. regression, decision trees, and neural networks
- Proposed projects and was holistically responsible for them from formatting data to choosing appropriate analysis
- Learned about predictive analytics algorithms and practices both from mentor and self-study from reference materials in office
- Implemented in SQL and R

Conferences/Workshops/Summer Schools

- **NITMB MathBio Convergence Conference - August 2025**
  - Poster presented: Modeling bias in decision-making attractor networks
- **Cognitive Foundations of Decision-Making - Mohammed VI Polytechnic University - July 2024**
  - Talk given: Modeling bias in decision-making circuits
- **Math + Neuroscience: Strengthening the Interplay Between Mathematics and Theory (Semester Program) - (Brown University) ICERM - Fall 2023**
  - Semester-long visitor for three workshops
- **Intro. Workshop on Computational Methods in Neuroscience - Campus Alberta Neuroscience - May 2023**
- **Penn State Neuroscience Institute Retreat - May 2023**
  - Poster presented: Modeling bias in decision-making attractor networks
- **COSYNE - March 2023**
- **Center for Neural Engineering Retreat - August 2022**
- **Society for Mathematical Biology Annual Conference - June 2021**

ADDITIONAL

Technical Skills: MATLAB, R, Python, C, CUDA, SQL, STATA
Languages: Fluent in English; Conversational Proficiency in Spanish, Hindi
Training: Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning (Coursera)